



**ANGLO-CHINESE JUNIOR COLLEGE**  
**Preliminary Examination 2021**

**BIOLOGY**  
**9744/01**

**HIGHER 2**

**15 September 2021**

**Paper 1 Multiple Choice**

**1 hour**

Additional Material: Multiple Choice Answer Sheet

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

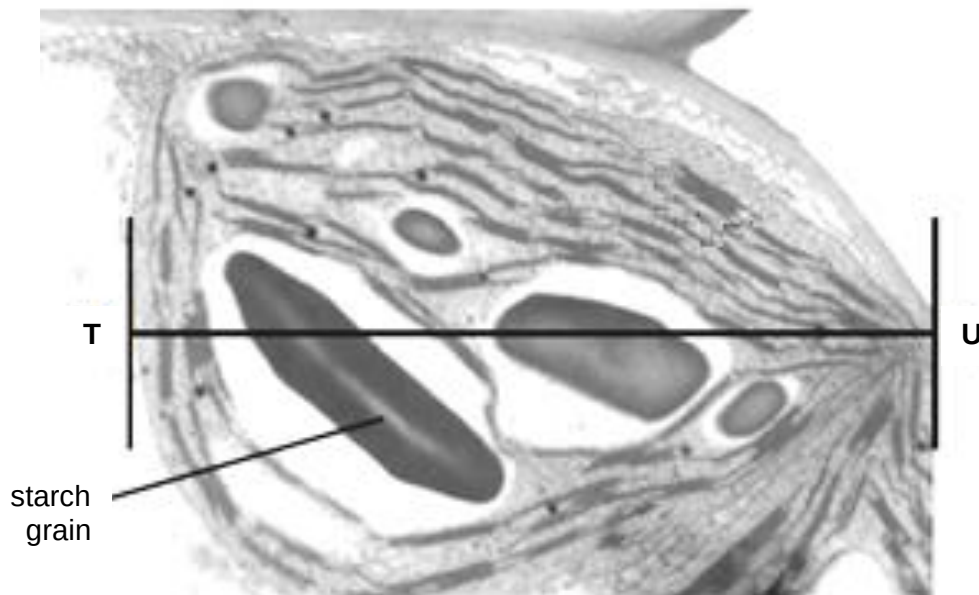
**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.  
Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **23** printed pages.

- 1 The electron micrograph shows part of a chloroplast in a plant cell. The actual length of the labelled starch grain in the chloroplast is  $2.2\text{ }\mu\text{m}$  long.



What is the actual length of this chloroplast indicated by the line between T and U?

- A** 48 mm                      **B**  $4.9\text{ }\mu\text{m}$                       **C** 107 mm                      **D**  $16.5\text{ }\mu\text{m}$

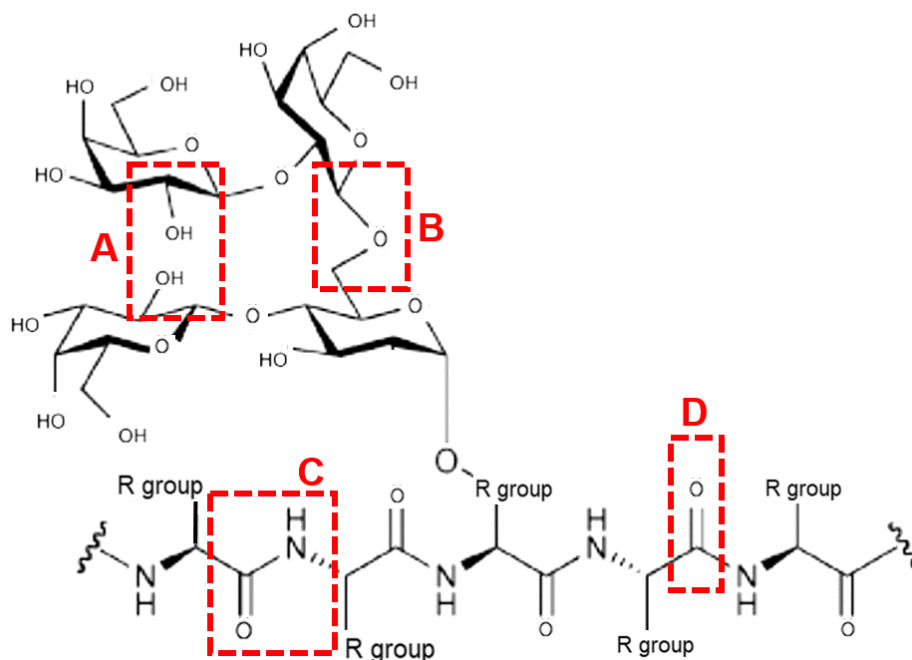
- 2 The majority of cells in the phloem tissue are companion cells and sieve tube elements. Companion cells load sucrose into sieve tube elements. A scientist isolated companion cells to investigate the mechanism that is coupled to the loading of sucrose into sieve tubes. He recorded these observations.

- 1 isolated companion cells become slightly negatively-charged compared to their surroundings
- 2 companion cells could decrease pH of the surrounding solution from 7.0 to 5.6
- 3 the pH inside the companion cells rose from 7.0 to 8.2
- 4 treatment with cyanide (which stops aerobic respiration) prevents changes in pH from occurring

Which conclusion can be drawn about the mechanism?

- A** Protein channels and protein carriers are involved in the mechanism of transport.  
**B** Polar and charged substances can be transported across cell surface membranes by facilitated diffusion and active transport.  
**C** Protons have moved into the companion cells by active transport with the expenditure of ATP.  
**D** Specific binding of ions to binding sites in protein carriers and conformational changes to the protein carriers have taken place.

- 3 The figure shows part of a glycoprotein with labelled boxes.



Which box, A, B, C or D, correctly labels a peptide bond?

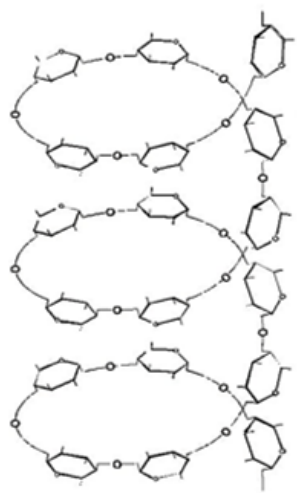
- 4 Haemoglobin and collagen serve important functions in mammals. A few statements describing haemoglobin and collagen are listed:

- 1 The secondary structures of both haemoglobin and collagen are stabilised by hydrogen bonds.
- 2 Polymerisation of both haemoglobin and collagen to form fibres allows for greater tensile strength.
- 3 The tertiary structures of both haemoglobin and collagen involve the further folding of the polypeptide chain containing  $\alpha$ -helices.
- 4 The primary structure of collagen is a repeating sequence while the primary structure of haemoglobin is non-repeating.
- 5 Covalent cross-links present in the quaternary structures of both haemoglobin and collagen maintain the 3D conformation of the molecules.

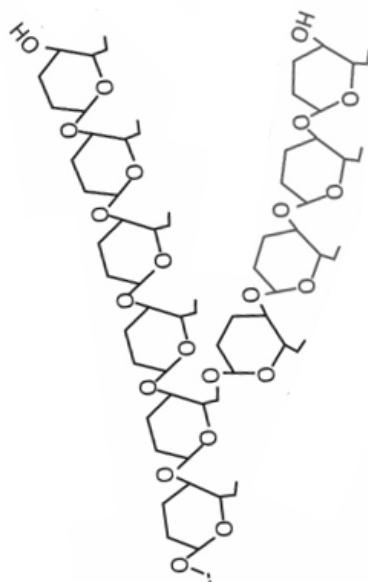
Which statements are correct?

- A** 1, 4 and 5      **B** 2, 3 and 4      **C** 1 and 4 only      **D** 3 and 5 only

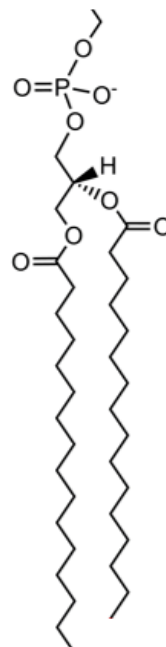
- 5 A student made observations of the structures and functions of the molecules W, X, Y and Z.



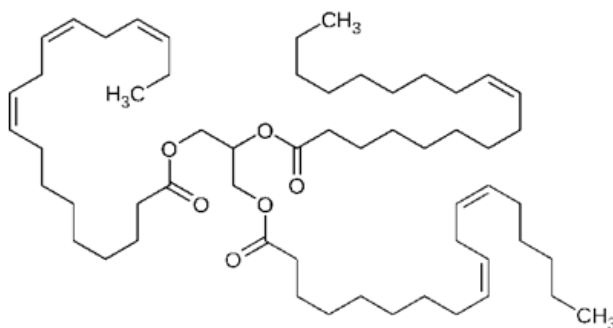
W



X



Y

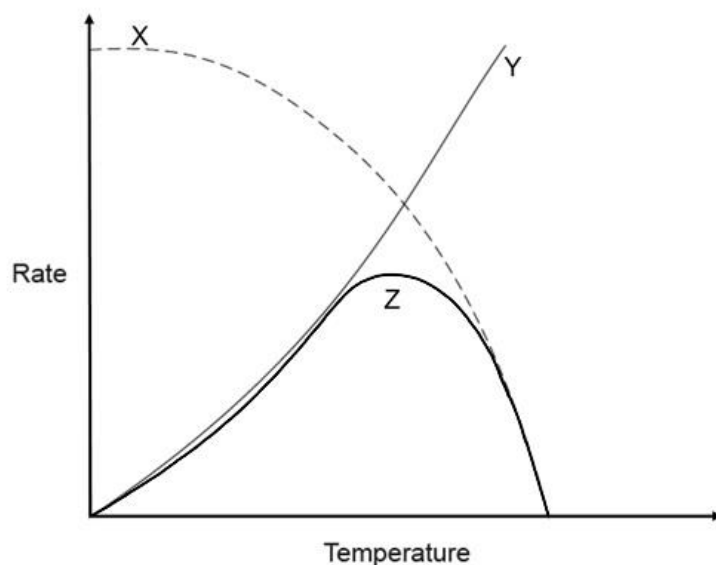


Z

Which row correctly states the structure and function of the specified molecule?

|          | molecule | structure               | function                                           |
|----------|----------|-------------------------|----------------------------------------------------|
| <b>A</b> | W        | helical and unbranched  | allows for the development of turgidity            |
| <b>B</b> | X        | helical and branched    | allows for quick release of respiratory substrates |
| <b>C</b> | Y        | long unsaturated chains | compartmentalises enzymes                          |
| <b>D</b> | Z        | branched and saturated  | maintains osmotic potential in the cells           |

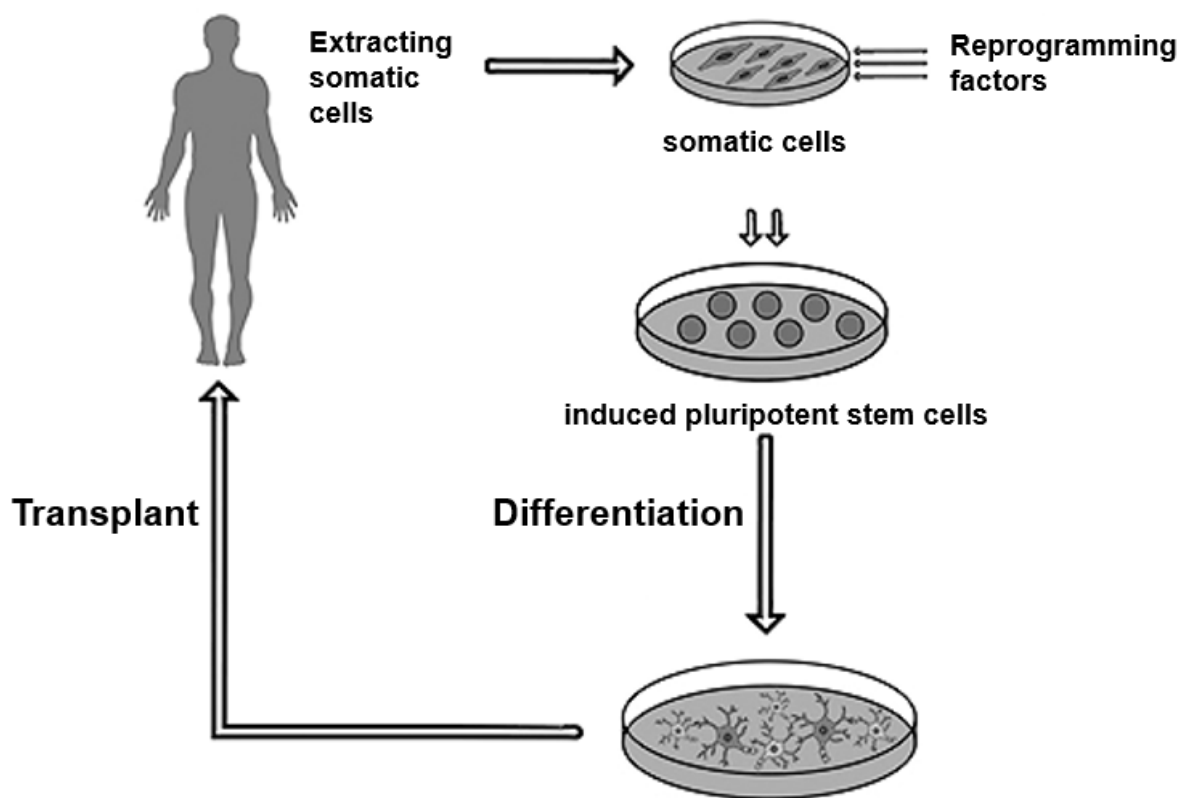
- 6 Investigating the rate of reaction of an enzyme at different temperatures gives rise to three different graphs, X, Y and Z.



What do graphs X and Y represent?

|          | X                                                                                  | Y                                                                                  |
|----------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>A</b> | Increasing temperature decreases the movement of enzymes                           | Increasing temperature affects the active site of the enzyme                       |
| <b>B</b> | Increasing temperature increases rate of collisions between enzymes and substrates | Increasing temperature increases the rate of reaction                              |
| <b>C</b> | Increasing temperature decreases the movement of enzymes                           | Increasing temperature increases the rate of reaction                              |
| <b>D</b> | Increasing temperature affects the active site of the enzyme                       | Increasing temperature increases rate of collisions between enzymes and substrates |

- 7 Stem cell research in regenerative medicine aims to generate healthy tissues for transplant into patients. The figure illustrates one particular stem cell therapy.



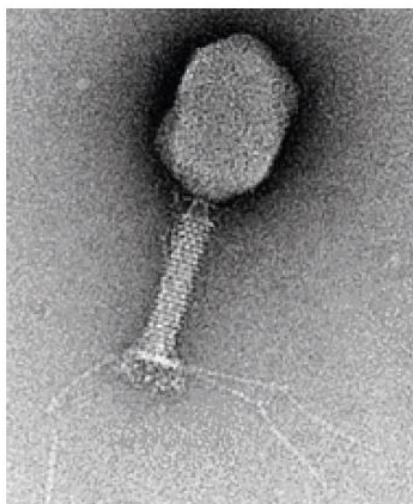
What is a possible ethical concern of this stem cell therapy?

- A The demand for tissues for transplant outweighs the supply.
  - B Some patients may develop cancer after the transplant procedure.
  - C Risk of exploitation of the patient to donate his cells.
  - D Harvesting the somatic cells from the skin is invasive to the donor.
- 8 These statements describe the variety of genomes present in viruses, prokaryotes and eukaryotes.
- 1 The genome is circular.
  - 2 The genome is compacted with proteins.
  - 3 The genome is made up of single-stranded RNA.
  - 4 The genome is made up of a few identical, segmented molecules.

How many statements correctly describe the prokaryotic genome?

- A 1
- B 2
- C 3
- D 4

- 9 Two scientists, Hershey and Chase, used the virus shown in the figure to investigate the process of transduction.



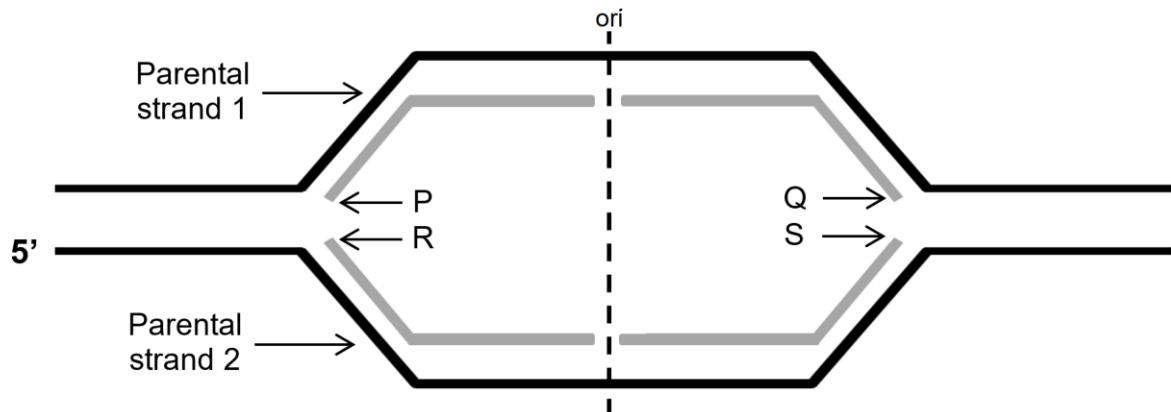
Sulfur atoms in proteins and phosphorus atoms in DNA of the virus were radioactively-labelled. The bacterial cells were infected with the radioactively-labelled virus and centrifugation of the bacterial culture is carried out. The cells will sink after centrifugation to form the pellet. The proportion of radioactive sulphur and phosphorus in the supernatant and pellet are recorded in the table.

| Sample source          | Supernatant | Pellet |
|------------------------|-------------|--------|
| Radioactive sulphur    | 82%         | 18%    |
| Radioactive phosphorus | 26%         | 74%    |

Which statement agrees with the results from Hershey and Chase's experiment?

- A Viruses infect bacterial cells with proteins.
- B Infection occurs when viral DNA was found within the bacterial cells.
- C Radioactively-labelled viral proteins was produced by the bacterial cells.
- D Neither the viral proteins nor viral DNA entered the bacterial cells during infection.

- 10** A simplified representation of a replication bubble is shown in the figure. The origin of replication (ori) is located at the middle of the DNA molecule. Parental strands 1 and 2 and the ends of the growing daughter strands P, Q, R and S are indicated.



Which ends will be shorter than the other ends of the resulting daughter strands when replication is complete?

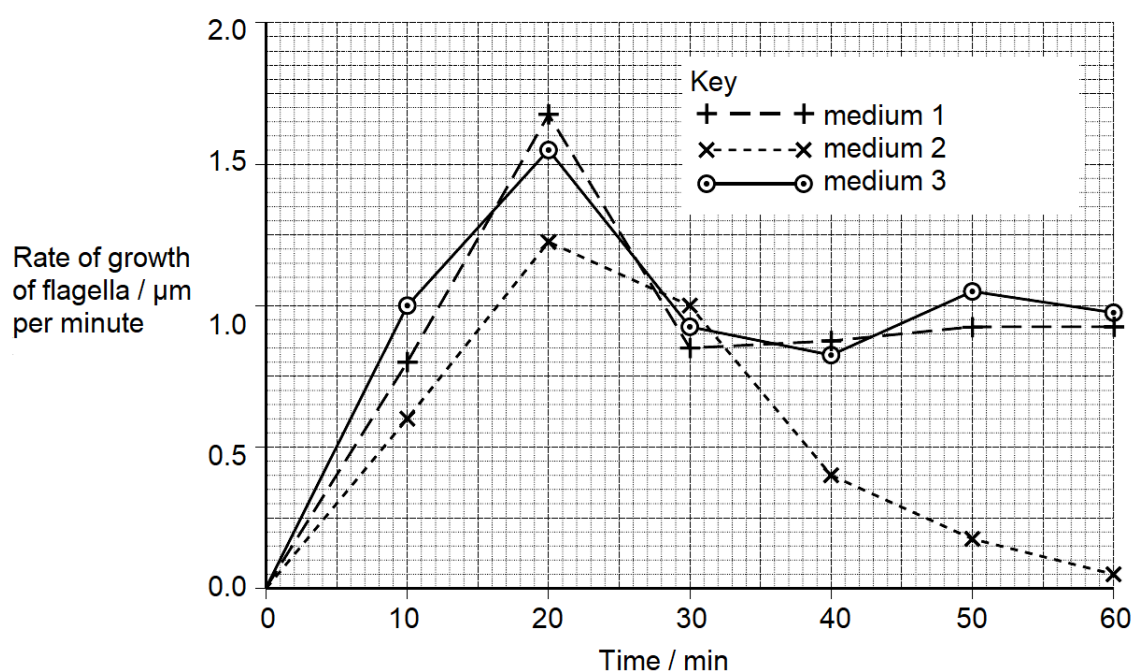
- A** P and R
- B** P and S
- C** Q and R
- D** Q and S



- 11** A group of researchers investigated the rate at which the flagella of a unicellular organism grew in three different types of media.

| medium | contents                                                                          |
|--------|-----------------------------------------------------------------------------------|
| 1      | contains actinomycin D, which prevents transcription by binding to guanine in DNA |
| 2      | contains puromycin, which prevents translation by attaching to ribosomes          |
| 3      | contains neither actinomycin D nor puromycin                                      |

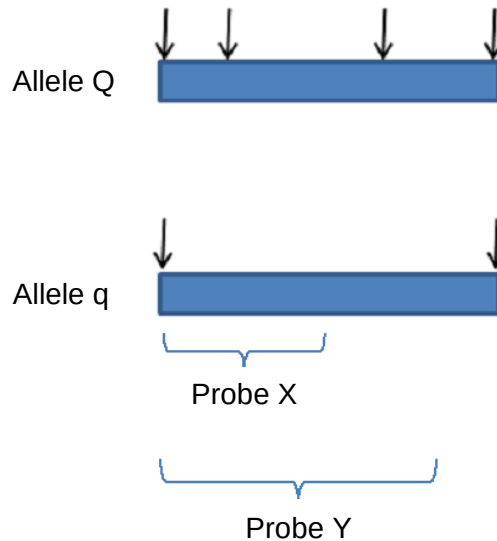
The graph shows the experimental results obtained by the researchers.



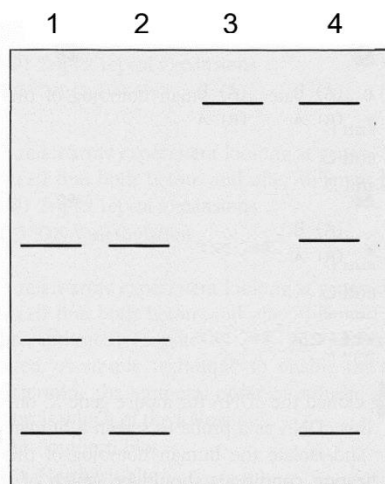
Which statement is the most likely explanation for the results obtained?

- A** Actinomycin D did not inhibit transcription to a great extent for 60 minutes as the graphs of medium 1 and 3 are similar.
- B** There was sufficient pre-existing mRNA for normal flagella growth for the organism in medium 1 throughout the experiment.
- C** There was a lag time of 30 minutes before puromycin inhibited translation in medium 2, causing a sharp decrease in the rate of flagella growth.
- D** There were sufficient pre-existing proteins for normal flagella growth for the organism in all media for the first 30 minutes only.

- 12** The figure shows two alleles of a gene implicated in a disease. The arrows indicate the sites at which a restriction enzyme will cleave the DNA while the curved brackets show the locations where probes X and Y will anneal.



Four DNA samples were digested with the restriction enzyme and separated by gel electrophoresis. Southern blotting was then carried out. Samples from homozygous individuals were treated with probe Y, while the samples from heterozygotes were treated with probe X. The combined results are shown in the autoradiogram.

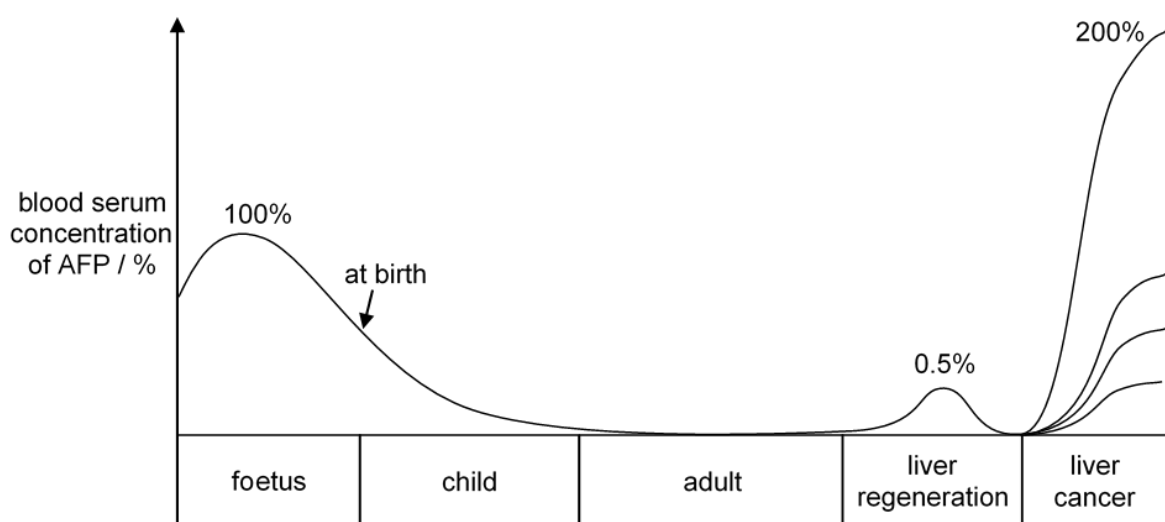


Which row correctly matches the genotype to the samples in the autoradiogram?

|          | Homozygous for allele Q | Homozygous for allele q | Heterozygous |
|----------|-------------------------|-------------------------|--------------|
| <b>A</b> | 3                       | 4                       | 1, 2         |
| <b>B</b> | 1, 2                    | 3                       | 4            |
| <b>C</b> | 2, 4                    | 3                       | 1, 2         |
| <b>D</b> | 1                       | 2, 4                    | 3            |

- 13 Which statement regarding post-transcriptional modification is **not** true?
- A The addition of a 5' modified guanosine cap is required for the mRNA to exit the nucleus into the cytoplasm via nuclear pores.
  - B Complexes of RNA and proteins recognise splice sites to excise introns and join exons together.
  - C Alternative RNA splicing allows the same type of cell to use the same gene to synthesise different proteins at different periods in the organism's life.
  - D The addition of a 3' poly-(A) tail prevents the degradation of mRNA by exonucleases.
- 14  $\alpha$ -fetoprotein (AFP) in blood serum is commonly used as a biomarker for the detection of liver cancer. The *AFP* gene is expressed only in the human foetus but not in a healthy adult.

The graph (not drawn to scale) shows the blood serum levels of AFP at various stages of life.

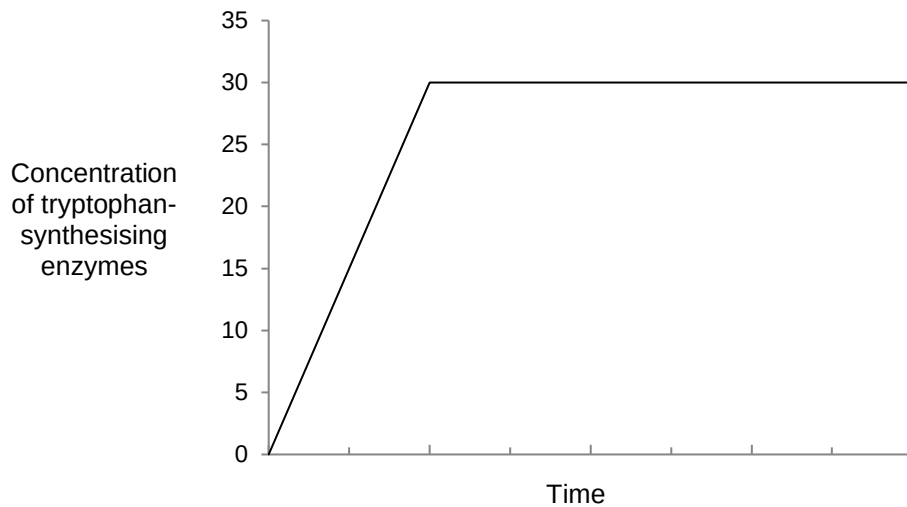


Which statements are correct?

- 1 The *AFP* gene is found in euchromatin and not heterochromatin in the nucleus of a foetal cell.
  - 2 Upon birth, deacetylation of lysine residues of histones could have occurred during chromatin remodelling.
  - 3 It may be difficult to distinguish early-stage liver cancer from liver tissue regeneration.
- A 1, 2 and 3
  - B 1 and 2 only
  - C 1 and 3 only
  - D 2 and 3 only

- 15 The *trp* operon comprises the promoter, operator and structural genes. Some of these components could become non-functional.

An investigation was carried out to determine which components are non-functional. The graph shows the concentration of tryptophan-synthesising enzymes in a bacterial cell containing the *trp* operon. The cell was transferred from a medium containing tryptophan to another without tryptophan at the start of the experiment.



key

- |   |                            |      |                                 |
|---|----------------------------|------|---------------------------------|
| + | : Functional component     | pro  | : promoter                      |
| - | : Non-functional component | o    | : operator                      |
|   |                            | sg   | : structural genes              |
|   |                            | trpR | : gene coding for trp repressor |

Which of the following combinations could lead to the graph above?

- A trpR<sup>-</sup> pro<sup>+</sup> o<sup>+</sup> sg<sup>+</sup>
- B trpR<sup>-</sup> pro<sup>+</sup> o<sup>-</sup> sg<sup>+</sup>
- C trpR<sup>+</sup> pro<sup>-</sup> o<sup>+</sup> sg<sup>-</sup>
- D trpR<sup>+</sup> pro<sup>+</sup> o<sup>+</sup> sg<sup>+</sup>

- 16** Some species of reptiles and fish can undergo parthenogenesis. This is a method of reproduction where the female parent can produce offspring without a male partner. In parthenogenesis, an unfertilised egg cell develops into an embryo. However, scientists are worried about the survival of these species in the long run.

Which statement explains the scientists' concern?

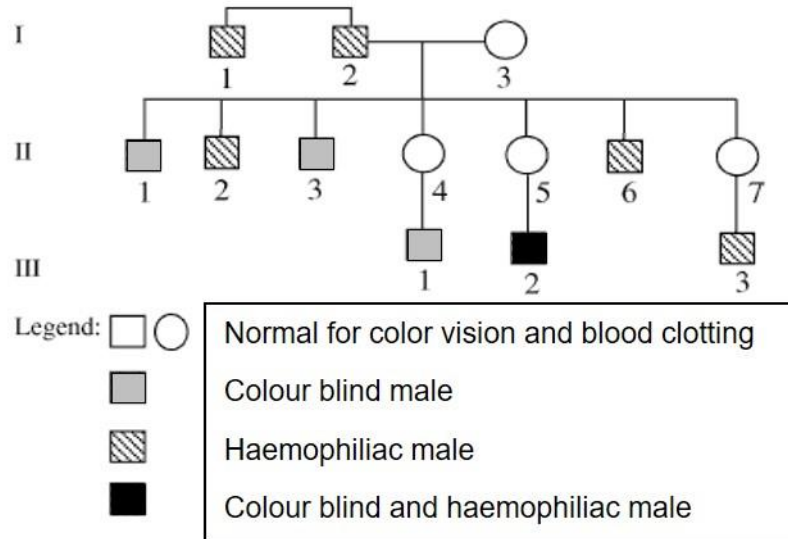
- A** Sexual reproduction, which is required for high number of offspring, cannot occur.
  - B** Population cannot resist environmental change due to low genetic variation.
  - C** Diploid individuals, which can adapt better, are not present in the population.
  - D** Gametes are not fertilised and can only divide for a finite number of times.
- 17** A test cross of some *Drosophila* fruit flies was expected to obtain a phenotypic ratio of 1:1:1:1. A chi-squared test was performed to test the significance of the difference between the observed and expected results. The chi-squared value obtained was 8.32. Based on the chi-squared table, a student concluded that the ratio of 1:1:1:1 is not valid.

| degrees of freedom | probability, <i>p</i> |      |      |      |       |       |       |
|--------------------|-----------------------|------|------|------|-------|-------|-------|
|                    | 0.50                  | 0.20 | 0.10 | 0.05 | 0.02  | 0.01  | 0.001 |
| 1                  | 0.46                  | 1.64 | 2.71 | 3.84 | 5.41  | 6.64  | 10.38 |
| 2                  | 1.39                  | 3.22 | 4.61 | 5.99 | 7.82  | 9.21  | 13.82 |
| 3                  | 2.37                  | 4.64 | 6.25 | 7.82 | 9.84  | 11.35 | 16.27 |
| 4                  | 3.36                  | 5.59 | 7.78 | 9.49 | 11.67 | 13.28 | 18.47 |

Which statement best explains the student's conclusion?

- A** There is less than 95% probability that the difference between observed and expected ratios is due to chance.
- B** There is less than 95% probability that the expected ratio is correct.
- C** There is less than 5% probability that the difference between observed and expected ratios is due to chance.
- D** There is less than 5% of the results fitting the expected ratio, suggesting a significant difference between the observed and expected ratios.

- 18 The pedigree traces the inheritance of two sex-linked genetic diseases in a family.

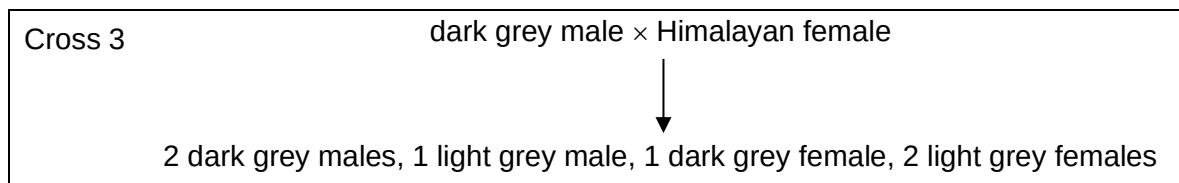
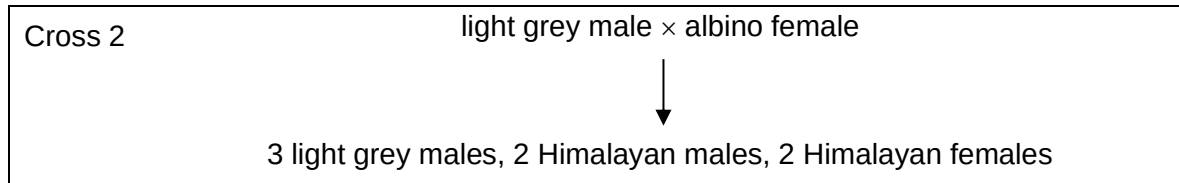
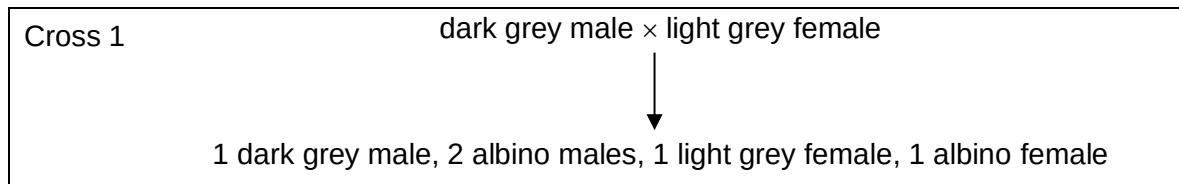


The fathers of individuals from generation III are normal for colour vision and blood clotting.

Which statement is **not** true?

- A Crossing over between the two genes may have resulted in the phenotype of III-2.
- B II-4 is a carrier of only one of the diseases.
- C II-7 is a carrier of both diseases.
- D The probability of III-3 having a colour blind child is 25%, assuming his wife is a carrier for colour blindness.

- 19 The figure shows the results of three crosses with rabbits of different fur colours.



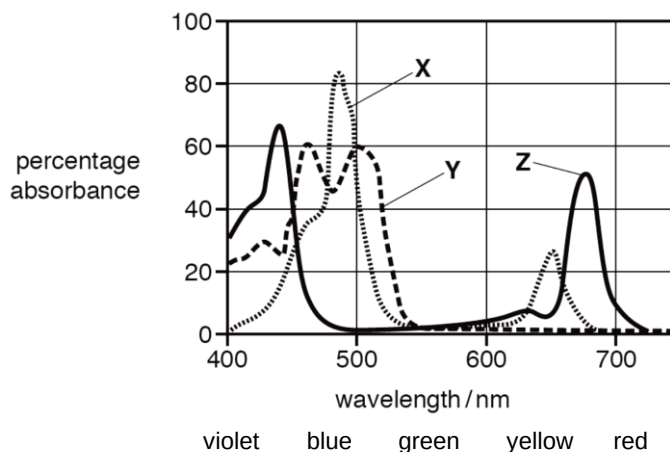
Which mode of inheritance explains the range of phenotypes shown by these crosses?

- A Sex linkage of gene coding for fur colour
- B Two genes, with a Mendelian mode of inheritance
- C Two genes, with one gene epistatic over the other
- D One gene, multiple alleles

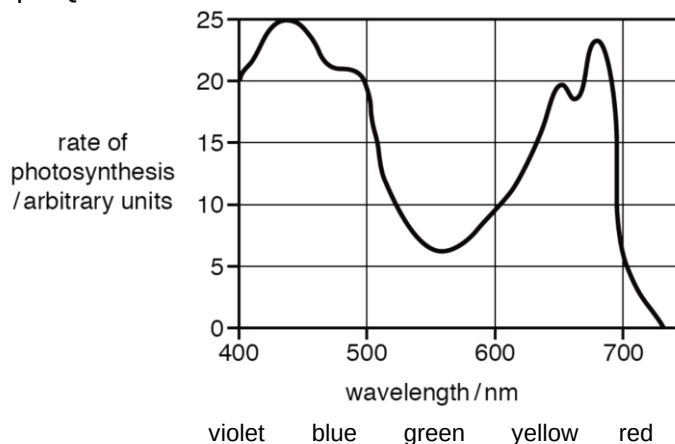
- 20** Graph P shows the absorption spectra of three types of photosynthetic pigment, X, Y and Z, extracted from the leaves of a flowering plant.

Graph Q shows the action spectrum of photosynthesis for the same plant.

Graph P



Graph Q



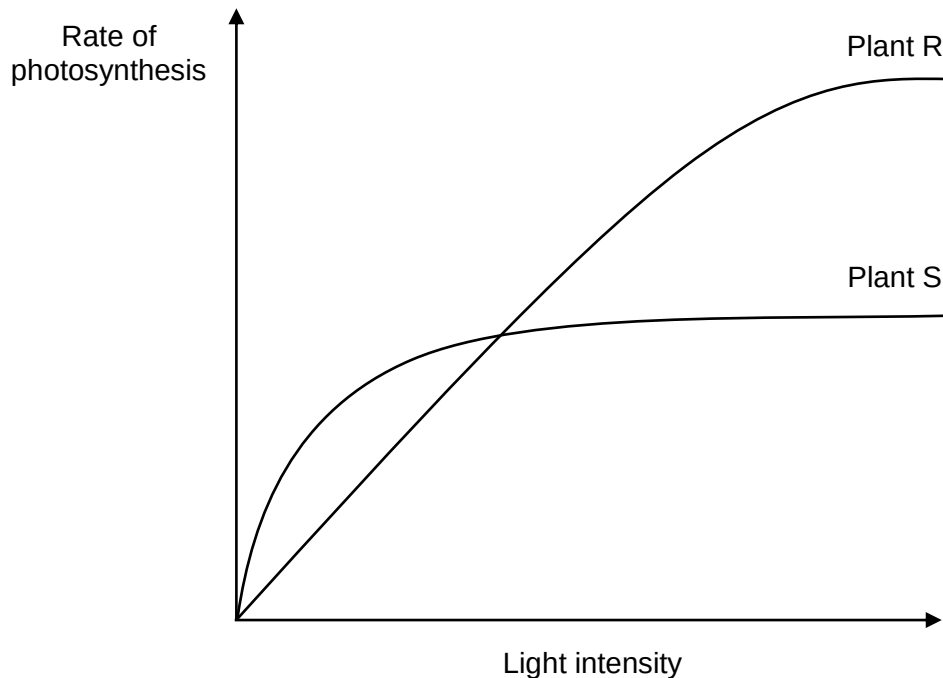
Which statement is a correct conclusion based on graphs P and Q?

- A** Pigment Y expands the ability of the plant to absorb light in the blue-green part of the spectrum but not the yellow-green part of the spectrum.
- B** Pigment Y is in Photosystem I since the plant has a low rate of photosynthesis at wavelength 700 nm.
- C** The poor absorption of green light by all three pigment types suggests that only cyclic photophosphorylation can occur.
- D** The rate of photosynthesis between the wavelengths 400 nm to 500 nm is from the cumulative absorption of light of pigments X and Z only.



- 21 Plant R can be found in abundance in an open field and plant S can be found in abundance in shady areas.

The graph shows the effect of increasing light intensity on the rate of photosynthesis of plants R and S.



Which statements are possible explanations of the rates of photosynthesis of the two plants?

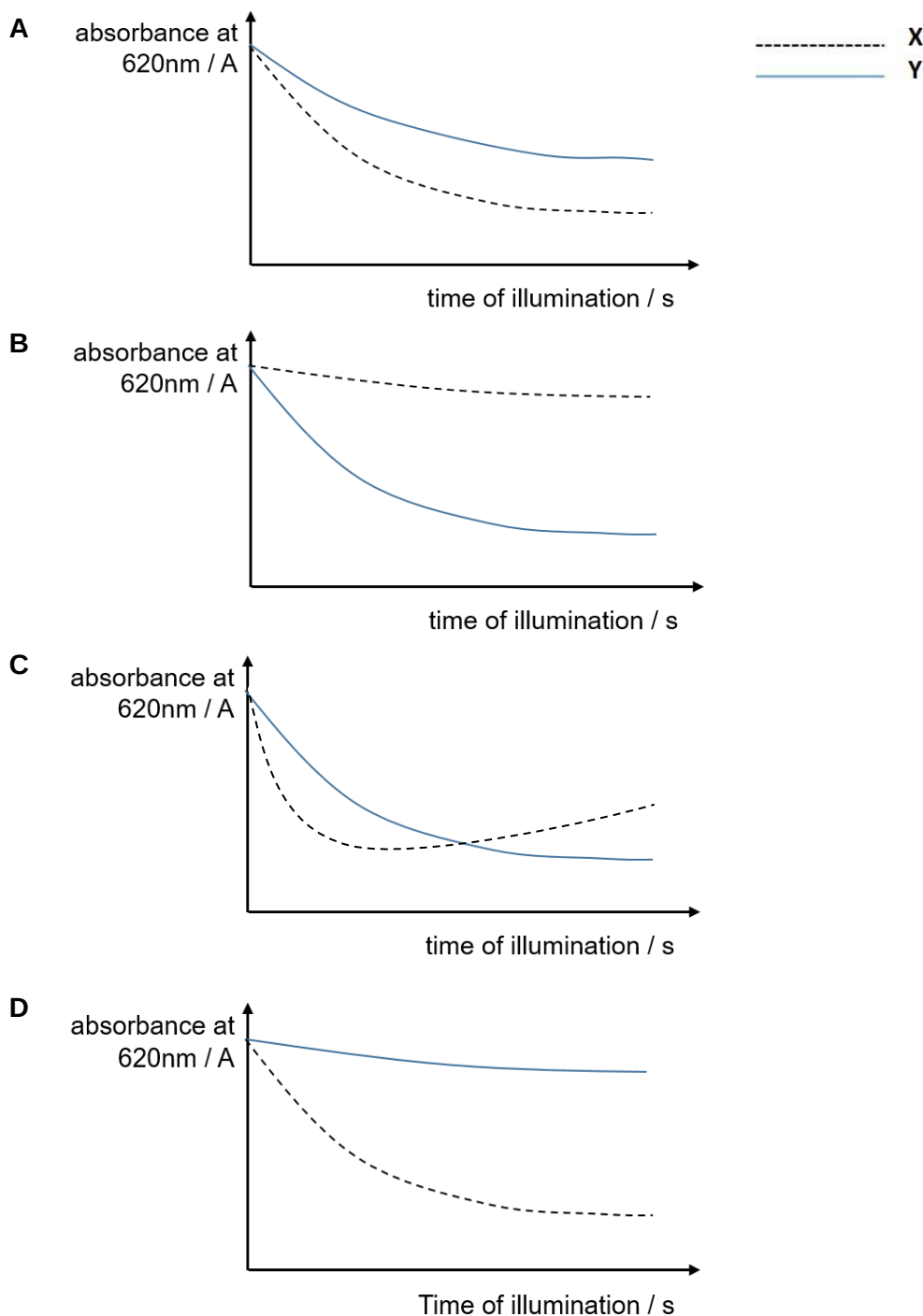
- 1 The rate of photosynthesis of both plants R and S at high light intensity is limited by temperature and carbon dioxide.
- 2 The graph for plant S shows a higher rate of photosynthesis at low light intensity because of a higher concentration of chlorophyll.
- 3 The rate of photosynthesis of plant S is higher at low light intensity due to its larger leaves for more light absorption.

- A** 1, 2 and 3  
**B** 1 and 2 only  
**C** 1 and 3 only  
**D** 2 and 3 only

- 22** DCPIP is a blue dye which acts as an electron acceptor and can be used to study the light-dependent reaction of photosynthesis. When blue DCPIP is reduced by NADP<sup>+</sup> reductase, it decolourises. The colour change is indicated by a drop in absorbance, measured using a colourimeter.

Two samples of leaf extracts, X and Y, were prepared. In X, inhibitors of NADP<sup>+</sup> reductase were added. These extracts were then mixed with DCPIP solution and exposed to light, and their absorbance recorded over time.

Which graphs correctly show the results of the above investigation?



- 23** Aerobic respiration involves two reactions, substrate-level phosphorylation and oxidative decarboxylation. These processes are labelled P and Q respectively in the table.

Which row is correct?

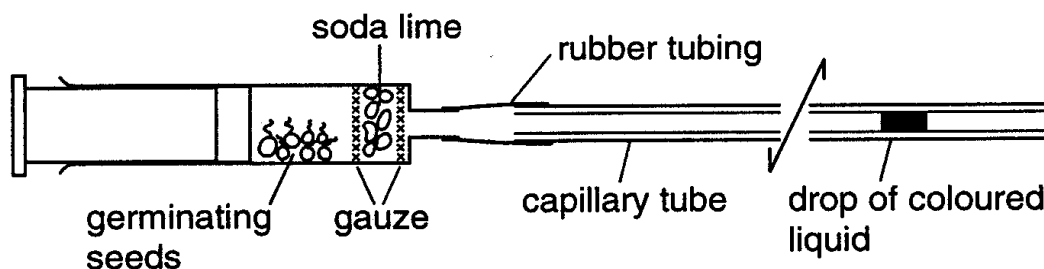
Key:

✓ involves

✗ does not involve

|          | glycolysis |   | link reaction |   | Krebs cycle |   |
|----------|------------|---|---------------|---|-------------|---|
|          | P          | Q | P             | Q | P           | Q |
| <b>A</b> | ✓          | ✗ | ✓             | ✓ | ✓           | ✓ |
| <b>B</b> | ✗          | ✗ | ✗             | ✓ | ✓           | ✓ |
| <b>C</b> | ✓          | ✗ | ✗             | ✓ | ✓           | ✓ |
| <b>D</b> | ✓          | ✗ | ✗             | ✓ | ✗           | ✓ |

- 24** The set-up investigates the effect of two temperatures, 25°C and 37°C, on the rate of respiration in germinating seeds. Soda lime has been placed inside the syringe to absorb carbon dioxide. The movement of the drop of coloured liquid can be observed and recorded to calculate the rate of respiration.



Which observation is correct?

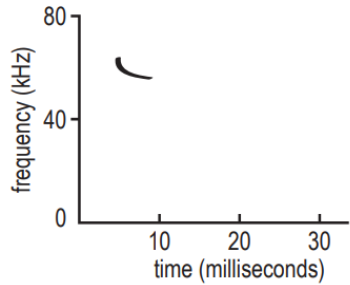
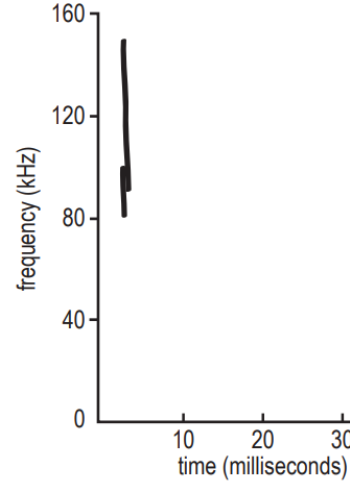
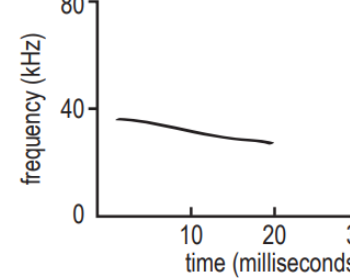
- A** The drop of coloured liquid moves towards the syringe at a faster rate at 25°C than at 37°C.
- B** The drop of coloured liquid moves away from the syringe at a faster rate at 25°C than at 37°C.
- C** The drop of coloured liquid moves towards the syringe at a faster rate at 37°C than at 25°C.
- D** The drop of coloured liquid moves away from the syringe at a faster rate at 37°C than at 25°C.

- 25** Which statement is true about a cell signalling pathway involving the decrease of glucose concentration to the set point?
- A** Activated protein kinase A phosphorylates and activates glycogen phosphorylase for glycogenolysis.
  - B** Dimerisation of receptor serves to activate tyrosine kinase to cross-phosphorylate the tyrosine residues on each other's cytoplasmic tails.
  - C** Phosphorylated receptor activates G-protein by replacing a molecule of GDP with GTP on the G-protein.
  - D** The ligand insulin is hydrophobic in nature.
- 26** The evolutionary success of mammals may be due to their well-developed sense of hearing which allow them to better locate food and avoid predators. This sense of hearing is partly due to three tiny bones in the middle ear, which scientists believe are developed from the small bones in the jaw of an early mammal. This early mammal later evolved into the three modern mammalian groups: monotremes, marsupials and placental mammals.

Which term describes the evolution of the three mammalian groups from the early mammal?

- A** Adaptive radiation
- B** Convergent evolution
- C** Microevolution
- D** Divergent evolution

- 27** Barro Colorado Island is a small island covered by tropical forest in Central America. Many bat species which live in the forest find their way around in darkness by emitting high-frequency sound pulses. These pulses bounce off obstacles and prey and allow them to judge the distance to an object. This behaviour is called echolocation. Features of three Barro Colorado Island species are described in the table.

| Species                  | Facial appearance of bat                                                            | Diet                      | Echolocation signal                                                                  |
|--------------------------|-------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------|
| Black myotis bat         |    | Insects                   |    |
| Mexican long-tongued bat |    | Nectar and pollen flowers |   |
| Velvety free-tailed bat  |  | Insects                   |  |

Which statement is true?

- A** Based on their dietary preferences, black myotis bat and velvety free-tailed bat do not share a recent common ancestor.
- B** The three species of bats can be classified as different species due to different facial appearances.
- C** Mexican long-tongued bat has a selective advantage because it has the longest echolocation signal.
- D** Evolution of the three species was an example of allopatric speciation.

- 28** COVID-19 is an acute respiratory illness that can be severe and is caused by a newly identified coronavirus named severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

Tests for the level of IgM and IgG antibodies for SARS-CoV-2 are carried out on newborn babies of six mothers with confirmed COVID-19 infection during pregnancy. When infected with a virus, humans initially produce IgM antibodies and then IgG antibodies. IgM antibodies do not cross the placenta.

The results are shown in the table. The normal level of IgM and IgG is below 10 AU/mL.

| antibody | antibody levels in infant blood samples (AU/mL) |          |          |          |          |          |
|----------|-------------------------------------------------|----------|----------|----------|----------|----------|
|          | infant 1                                        | infant 2 | infant 3 | infant 4 | infant 5 | infant 6 |
| IgM      | 39.6                                            | 16.25    | 3.79     | 1.9      | 0.96     | 0.16     |
| IgG      | 125.5                                           | 113.91   | 75.49    | 73.19    | 51.38    | 7.25     |

Which statement is a possible conclusion based on the information provided?

- A** The mother of infant 6 has no immunity to SARS-CoV-2.
  - B** Infant 1 to 5 will have about two years of passive immunity to SARS-CoV-2.
  - C** Infant 1 and 2 received IgM antibodies to SARS-CoV-2 from their mother.
  - D** Infant 1 and 2 were exposed to SARS-CoV-2 during fetal development.
- 29** Changes in the immunoglobulin genes of a B cell can occur from its development until its activation.

Which statement is true?

- A** Immunoglobulin genes can only be found in B cells.
- B** DNA rearrangements of the different V, D or J gene segments coding for the antibody chains occur in immature B cells.
- C** Somatic hypermutation changes the V domains of antigen such that they have a higher affinity to the antigen receptors on B cells.
- D** Class switching allows activated B cells to switch among different classes of antibodies reversibly depending on the needs of the body.

- 30** The spider-orchid only flowers for a relatively brief time. Its flower resembles a spider and emits a scent that mimics that of a female bee, tricking male bees into attempting to mate with the flowers, thereby allowing the plant's pollination.

The bees emerge in spring after hibernation and take flight, with the males flying earlier than the females.

Global warming has resulted in temperatures in the spring months to be higher than before, affecting the interaction between the spider-orchid and the male bees.

Which statements are possible concerns of higher temperatures in spring?

- 1 Female bees are emerging from hibernation earlier than usual, competing with the orchid flower and preventing it from being pollinated by the male bees.
- 2 The spider-orchid flowers earlier, resulting in desynchronisation between the pollination process and the hatching of the bee larvae.
- 3 The hatching process of the bee larvae is slower as some of the enzymes involved are inactivated at higher temperatures, resulting in the later emergence of the bees.

- A** 1, 2 and 3  
**B** 1 and 2 only  
**C** 1 and 3 only  
**D** 2 and 3 only

**-- End of Paper --**